

# A systematic review of state-of-the-art Laryngeal Electromyography in pediatric populations and meta-analysis of recovery prediction in vocal cord paralysis

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## ABSTRACT

**Objectives:** To provide a comprehensive evaluation of state-of-art Laryngeal Electromyography (LEMG) in pediatric population and quantitatively assess its prognostic utility in vocal cord paralysis (VCP).

**Methods:** The systematic review and meta-analysis were carried out and reported in accordance with Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines. A comprehensive literature search was performed across PubMed, Scopus, and Google Scholar. Eligible studies included case reports, case series, and observational studies. Data were extracted qualitatively and pooled using a random-effects model to evaluate the prognostic utility of LEMG in predicting VCP recovery.

**Results:** A total of 33 studies were included in the qualitative analysis, and five studies for meta-analysis. The mean age of patients in studies included for qualitative analysis was  $6.77 \pm 5.73$  years, with an age range from day 1 of life to 18 years. In contrast, the studies included in the meta-analysis enrolled children ranging from day 1 of life to 14 years, with a mean age of  $2.01 \pm 1.03$  years.

A random-effects meta-analysis of sensitivity of three studies yielded a pooled sensitivity of 82% (95% CI: 51–96%). A meta-analysis of Positive Predictive Value (PPV) of two studies, showed a pooled PPV of 75.0% (95% CI: 27.0% to 96.0%). Cox Proportional Hazards regression analysis showed that patients with Motor Unit Action Potential (MUAP) present had 5.56 times higher likelihood of recovery than those without MUAP (Hazard Ratio = 5.56, 95% CI: 1.65–18.71,  $p = 0.00558$ ). The Kaplan Meier (KM) curve analysis demonstrated patients with MUAP exhibited higher chances of recovery over time compared to those without MUAP ( $p = 0.00094$ ).

**Conclusion:** LEMG in pediatric laryngology serves a valuable diagnostic, therapeutic, and prognostic adjunct with diverse applications. It shows moderate accuracy in predicting favorable recovery in cases of VCP when normal MUAPs are present.

## 1. Introduction

Pediatric airway pathologies, particularly the vocal cord paralysis (VCP), present unique challenges due to the complex neural innervation extending from the premotor cortex to the intrinsic laryngeal muscles [1]. With a myriad of etiologies ranging from idiopathic causes to infections and trauma, accurately identifying the underlying pathology, prognosticating outcomes, and formulating the appropriate management plan remains a significant challenge for surgeons.

Laryngeal electromyography (LEMG), first introduced in 1944, has evolved into a versatile diagnostic tool in laryngology, widely employed for diagnosing, prognosticating, and serving as an adjunct to therapeutic

procedures in various laryngeal disorders [2]. It utilizes electrodes to capture the electrical activity of laryngeal muscles during voluntary movements, with the recorded signals analyzed by a neurophysiologist to draw clinically relevant conclusions. Despite its potential, LEMG's application is still limited to a handful of centers due to the need for specialized expertise, and high-quality studies with control groups are lacking. LEMG has been used widely in the adult population.

A 2012 meta-analysis by Rickert et al., which predominantly involved the studies in an adult population, showed the predictability of LEMG for non-recovery is better than for the recovery of VCP with a Positive Predictive Value (PPV) of 90.9% [3]. Similarly, a review by Sataloff et al. showed weak yet supporting evidence on use of LEMG in

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injecting Botulinum Toxin in spasmodic dysphonia [4]. While LEMG's utility in adults is well-documented, its role in pediatric laryngology remains underexplored. Furthermore, its potential to predict recovery in VCP, a condition that is not uncommon but poses significant uncertainty for patients and parents, remains unclear. This uncertainty can impact management decisions and parental counseling.

This systematic review and meta-analysis aim to provide a comprehensive evaluation of the state-of-the-art LEMG in pediatric patients and to quantitatively assess its prognostic capability in pediatric VCP. By synthesizing available evidence, this study seeks to establish a clearer understanding of LEMG's diagnostic and prognostic roles in pediatric laryngology, potentially standardizing its use and interpretation in this population.

## 2. Methodology

This systematic review and meta-analysis were conducted adhering to the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines [5]. A comprehensive literature search was performed across three databases: PubMed, Scopus, and Google Scholar. The detailed search strategy is provided in the Supplementary Material. Duplicate studies were identified and removed using Covidence software. The study selection process is illustrated in Fig. 1.

### 2.1. Inclusion criteria

#### 2.1.1. Study types

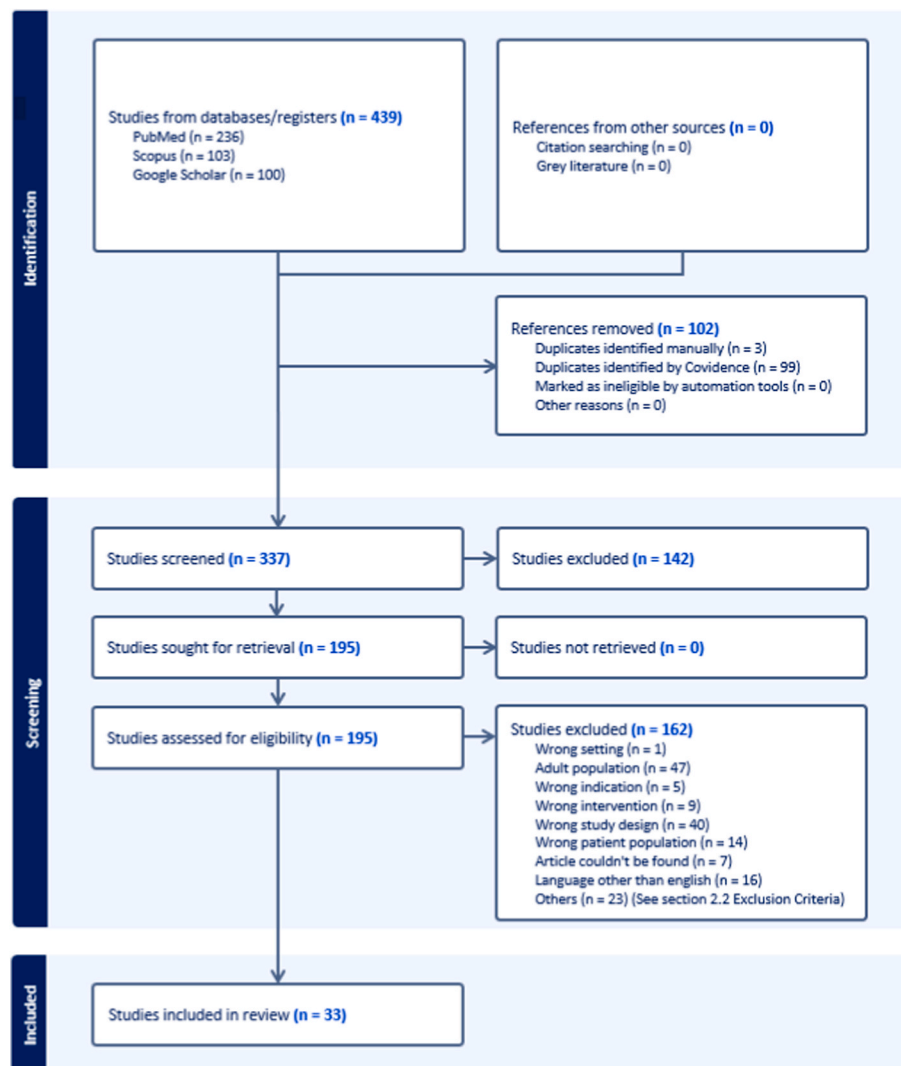
We included retrospective or prospective case series, case reports, and experimental studies reporting the use of LEMG in pediatric populations until June 30th, 2025 for qualitative analysis. For meta-analysis, studies reporting the prognostic utility of LEMG in predicting VCP recovery were included.

#### 2.1.2. Population

The study population were pediatric patients ( $\leq 18$  years) who underwent LEMG for any indication. For meta-analysis, we included studies reporting outcomes of pediatric patients with VCP having a minimum follow-up duration of 12 months for non-recovered cases. In studies reporting mixed populations, those were included if selective pediatric data were retrievable.

#### 2.1.3. Intervention

Studies employing LEMG with monopolar, bipolar, or surface electrodes were included. For the meta-analysis, only LEMG studies reporting outcomes in terms of motor unit action potential (MUAP)s without the use of muscle relaxants were considered.



**Fig. 1.** Flowchart for the article selection process of the systematic review and meta-analysis, showing the number of studies identified, screened, included, and excluded with reasons.

#### 2.1.4. Outcomes

For qualitative analysis, any study reporting outcomes after LEMG was included and for meta-analysis, studies had to report VCP recovery outcomes with a minimum follow-up duration of 12 months in cases without recovery.

#### 2.2. Exclusion criteria

Studies published in languages other than English, review articles, animal studies, or studies exclusively on adult populations, studies without clear indications for LEMG or lacking outcome data and studies with follow-up duration <12 months in cases with unrecovered VCP were excluded.

#### 2.3. Study selection and data extraction

Initial title and abstract screening were independently conducted by four authors (MP, SKC, SR, and SD). Full-text screening was completed in two rounds, with the first round by the same authors and the second by BRG for final verification. Discrepancies were resolved through consensus, consulting BRG when needed.

Data extraction was performed using a pre-designed Microsoft Excel template, capturing the following:

Qualitative Analysis: Study type, publication year, number and age range of patients, underlying pathology, LEMG indications, type of electrodes used, and summary of outcomes.

Meta-Analysis: Presence or absence of MUAP, nature of volitional and spontaneous activities, follow-up duration, and VCP recovery outcomes.

Classification of Cases for Meta-Analysis.

Cases were categorized for diagnostic analysis as follows:

True Positive (TP): VCP with present MUAP, with recovery.

False Negative (FN): VCP with absent MUAP or the presence of spontaneous activity, with recovery.

False Positive (FP): VCP with present MUAP, without recovery within 12 months.

True Negative (TN): VCP with absent MUAP and/or presence of spontaneous activity, without recovery within 12 months.

The term “spontaneous activity” refers to pathologic electromyographic discharges at rest (e.g., fibrillation potentials or positive sharp waves), which indicate ongoing denervation. These differ from MUAPs, which represent voluntary, organized motor unit recruitment during attempted phonation and reflect functional reinnervation.

LEMG was considered “positive” when voluntary MUAP recruitment was present without pathologic spontaneous activity, indicating favorable prognosis. LEMG was considered “negative” when voluntary MUAPs were absent and/or spontaneous denervation activity was present, suggesting poor prognosis.

TP were cases in which LEMG predicted recovery and recovery occurred. FP were cases in which LEMG predicted recovery but no recovery occurred. FN were cases in which LEMG predicted non-recovery but recovery occurred. TN were cases in which LEMG predicted non-recovery and no recovery occurred.

In cases of serial LEMG, the earliest EMG was considered for recovery prediction. Recovery was defined as normal or sufficient vocal cord mobility on follow-up laryngoscopic examination, additionally reflected by sufficient improvement in airway function to permit safe decannulation in patients who underwent tracheostomy.

#### 2.4. Quality assessment

Study quality was assessed using the Newcastle–Ottawa Scale (NOS) [6] for observational studies and the Joanna Briggs Institute (JBI) Critical Appraisal Checklists for case reports and case series [7]. Independent scoring was performed by MP, SKC, SR, and SD. For NOS, scores of 7–9 indicated low risk of bias, 4–6 medium risk, and <4 high risk. For

the JBI checklist for case reports, scores of 7–8 denoted low risk, 5–6 medium risk, and <4 high risk. Similarly, case series assessed with the JBI checklist were classified as low (8–10 “yes” responses), medium (5–7), or high risk of bias (<5). The tables of the quality assessment have been provided in the supplementary material.

#### 2.5. Statistical analysis

This meta-analysis aimed to evaluate the diagnostic performance of LEMG in predicting VCP recovery, specifically focusing on sensitivity and PPV. Although our initial analysis plan included sensitivity, specificity, PPV, and Negative Predictive Value (NPV), we ultimately limited our analysis to sensitivity and PPV based on the availability and completeness of diagnostic data across included studies. Diagnostic Odds Ratio (DOR) was calculated only for studies with complete  $2 \times 2$  diagnostic data (TP, FP, FN, TN). Hazard ratio (HR) and Kaplan Meier (KM) Survival Curve analysis were carried out to account for time-dependent recovery outcomes, providing a dynamic measure of recovery probability over time.

All analyses were performed using RStudio Cloud (Posit Cloud version 4.3.0) with the ‘meta,’ ‘mada,’ and ‘ggplot2’ packages. A random-effects model (DerSimonian-Laird method) was used for pooling sensitivity and PPV due to expected clinical and methodological heterogeneity among studies.

### 3. Results

A total of 33 studies were included in the qualitative analysis, with five studies selected for meta-analysis (Fig. 1). The study population comprised children ranging from the first day of life to 18 years (Mean age:  $6.77y \pm 5.73$  years). The age range of children included in the studies selected for the meta-analysis was from day 1 of life to 14 years (Mean age:  $2.01 \pm 1.03$  years). Mean ages were not reported in all studies, and were derived from available case data where possible. All studies were case reports or case series (retrospective and prospective). LEMG electrodes used included either surface (endotracheal, esophageal or skin) or intramuscular.

The indications for LEMG in the pediatric population were broadly categorized into three groups for clarity:

1. **Diagnostic and Prognostic Applications:** This category primarily focused on confirming the presence of VCP and distinguishing it from other causes of vocal fold immobility (VFI), such as cricoarytenoid joint dislocation and posterior glottic stenosis (PGS). Additionally, it was used to predict the likelihood of vocal cord recovery based on EMG findings. A total of 17 studies were included in this group, with the findings summarized in Table 1 [8–24].
2. **Anatomical Mapping and Therapeutic Support:** LEMG was used for mapping or identifying the recurrent laryngeal nerve (RLN) or vagus nerve during procedures where these nerves were at risk, such as thyroid surgery, intracranial tumor resection, and vagus nerve stimulation (for intractable epilepsy). It also served as an adjunct in therapeutic decision-making, including laryngeal reinnervation, and as a tool for providing biofeedback during voice therapy. A total of 14 studies were included in this category, with the results detailed in Table 2 [25–38].
3. **Physiological Studies:** These studies explored the physiological and developmental aspects of respiratory function, specifically examining the phasic activity of the diaphragm and vocal cords, including the mechanisms of expiratory braking. Two studies were included in this group, with their findings presented in Table 3 [39,40].

A random-effects meta-analysis of sensitivity carried out across three studies: Maturo et al. [23], Giotakis et al. [18], Gartlan et al. [21], yielded a pooled sensitivity of 82% (95% Confidence Interval (CI): 51–96%) (Fig. 2(a)). The sensitivities from the individual studies ranged

**Table 1**  
Pediatric LEMG for diagnostic and prognostic indications.

| Author                            | Cases (n) | Age             | Pathology                                                          | Electrodes       | Summary of findings                                                                                                                                                                                                                                                                                                                              | Risk of Bias |
|-----------------------------------|-----------|-----------------|--------------------------------------------------------------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| Carrat et al., 2000 [8]           | 1         | 3 y             | Interarytenoid adhesion                                            | NM               | Presence of normal innervation.                                                                                                                                                                                                                                                                                                                  | Low          |
| Aragon-Ramos et al., 2022 [9]     | 10        | 5 m-17 y        | VFI/hypomobility                                                   | IM (TA)          | 40% cases had pathologic finding in LEMG. LEMG determined management plan in 60% cases and helped identification of PGS in 3 cases with bilateral VFI                                                                                                                                                                                            | Medium       |
| Wohl et al., 2001 [10]            | 30        | 49 d-17 y       | VFI, laryngotracheal abnormalities with or without VFI             | IM (TA)          | 83 % cases had correlation between direct laryngoscopy and LEMG. LEMG provided guidance for clinical management and consultation in 23.3% cases.                                                                                                                                                                                                 | Low          |
| Jacobs and Finkel, 2002 [11]      | 8         | 6 m – 5 y       | VFI                                                                | IM (TA, PCA)     | 37.5% cases with VFI had normal LEMG finding suggesting PGS and 50 % cases showed synkinetic activity on LEMG. LEMG helped in differentiating paralysis from fixation.                                                                                                                                                                           | Low          |
| Woo and Arnadia et al., 1992 [12] | 2         | 1.2 y and 1.8 y | VFI                                                                | IM (TA)          | LEMG was able to exclude VCP as the cause of VFI in two cases based on the presence of normal single motor unit potential                                                                                                                                                                                                                        | Medium       |
| Mallon et al., 2014 [13]          | 5         | 9-17 y          | Arytenoid dislocation                                              | IM (TA, PCA, CT) | Two cases had normal LEMG so underwent corrective procedure while three patients showed markedly reduced recruitment patterns suggesting associated RLN nerve injury                                                                                                                                                                             | Medium       |
| Koch et al., 1987 [14]            | 12        | 1m – 18 y       | VFI                                                                | IM (PCA)         | LEMG was useful in differentiating mechanical and paralytic VFI                                                                                                                                                                                                                                                                                  | Medium       |
| Ysunza et al., 2007 [15]          | 59        | 11-16 y         | Unilateral VCP (25), Arytenoid dislocation (9), Control group (25) | IM (TA, CT)      | LEMG had sensitivity of 100% and specificity of 92% in diagnosing VCP                                                                                                                                                                                                                                                                            | Low          |
| King et al., 2011 [16]            | 4         | <1 y            | Bilateral and Unilateral VCP                                       | IM (TA)          | LEMG was used to confirm recovery of VCP in all four cases                                                                                                                                                                                                                                                                                       | Medium       |
| Scott et al., 2011 [17]           | 3         | 2 wk to 11 m    | Bilateral or Unilateral VFI in cases with Arthrogryposis           | IM (TA)          | LEMG showed MUAP suggesting immobility to CA fixation                                                                                                                                                                                                                                                                                            | Medium       |
| Giotakis & Potosching, 2020 [18]  | 5         | <13 y           | Congenital Bilateral VCP                                           | IM (TA)          | All cases recovered over time and showed volitional activity in LEMG prior to recovery except in one case.                                                                                                                                                                                                                                       | Medium       |
| Scott et al., 2008 [19]           | 10        | 2 wk 14 y       | VCP                                                                | IM (TA)          | LEMG accurately predicted recovery of vocal fold paresis in 50% of the patients.                                                                                                                                                                                                                                                                 | Medium       |
| Berkowitz et al., 2009 [20]       | 2         | 55 d & 53 m     | Bilateral VCP                                                      | IM (TA, PCA)     | EMG findings showed that normal phasic activity; TA active during expiration and PCA during inspiration, was linked to good recovery, as seen in one child, while another child with aberrant PCA activity during expiration did not recover, suggesting that the presence and pattern of phasic activity may predict prognosis in bilateral VFP | Low          |
| Gartlan et al., 1993 [21]         | 4         | 41 wk to 14 y   | Unilateral or Bilateral VCP                                        | IM (TA)          | Patients with documented SMU activity (50%) showed favorable vocal cord recovery and those without SMU activity (50%) eventually required tracheostomy                                                                                                                                                                                           | Low          |
| Berkowitz RG, 1996 [22]           | 4         | 3 wk to 2.7 m   | Bilateral VCP                                                      | IM (TA, PCA)     | Presence of motor unit phasic bursts were not associated with recovery in all cases when followed up to an average of 3.7 years                                                                                                                                                                                                                  | Medium       |
| Maturo et al., 2011 [23]          | 25        | 14 d to 7 y     | Bilateral and Unilateral VFI                                       | IM (TA)          | All cases with VFI showing no MUAP or abnormal MUAP didn't show signs of recovery on at least 12 months of follow-up                                                                                                                                                                                                                             | Medium       |
| Reverdin et al., 2014 [24]        | 1         | 6.5 y           | Congenital central hypoventilation syndrome (CCHVS)                | IM (TA)          | LEMG detected vocal fold contractions synchronized with diaphragmatic pacing which was initially perceived as VCP in case of CCHVS                                                                                                                                                                                                               | Low          |

CCHVS, Congenital Central Hypoventilation Syndrome; CT, Cricothyroid; d, days; EMG, Electromyography; IM, Intramuscular; LEMG, Laryngeal Electromyography; m/mo, months; MUAP, Motor Unit Action Potential; NM, Not Mentioned; PCA, Posterior Cricoarytenoid; PGS, Posterior Glottic Stenosis; RLN, Recurrent Laryngeal Nerve; SMU, Single Motor Unit; TA, Thyroarytenoid; VCP, Vocal Cord Paralysis; VFI, Vocal Fold Immobility; wk, weeks; y, years.

from 68% to 96. Despite of low heterogeneity ( $I^2 = 24.3\%$ ,  $p = 0.267$ ), we applied a random-effects model to account for potential unmeasured clinical and methodological variation. For studies reporting perfect sensitivity (100%), a continuity correction of 0.5 was applied to all  $2 \times 2$  table cells to allow valid variance estimation. Sensitivity values were analyzed using the Logit Transformation, which provides stable estimation for proportions near 0 or 1. The relevant study-level data are provided in the supplementary material.

Similarly, meta-analysis of PPV of two studies, Maturo et al. [23] and Berkowitz et al. [20], showed a pooled PPV of 75.0% (95% CI: 27.0% to 96.0%)(Fig. 2(b)). Individual study PPVs ranged from 50 % to 89 %, with moderate to high between-study heterogeneity ( $I^2 = 65.6\%$ ,  $p = 0.0882$ ). Continuity correction (0.5) was also applied to ensure stable estimates.

The DOR for MUAP in predicting recovery was calculated for Maturo et al., the only study with complete  $2 \times 2$  diagnostic data. After applying a standard continuity correction to account for the absence of FNs, the DOR was 325 (95% CI: 12.2–8624.2). The DOR was 325 (95% CI: 12.2–8624.2), indicating a strong association between MUAP presence

and recovery. However, the wide CI reflects substantial uncertainty, likely due to the small sample size and zero FNs.

Cox Proportional Hazards regression analysis was conducted to evaluate the association between MUAP presence and recovery in patients with VCP which included four studies: Maturo et al. [23], Berkowitz et al. [20], Berkowitz RG [22] and Giotakis et al. [18] (Fig. 3(a)). Across these four studies, there were a total of 36 patients of which 18 had bilateral VCP (11 congenital, four central, and three idiopathic), and 18 had unilateral VCP, predominantly following surgery. Tracheostomy status was reported in only 10 cases. Among patients with bilateral VCP, two had associated craniofacial anomalies and developmental delay, and one had a history of repaired tracheoesophageal fistula with tracheobronchomalacia. Other comorbidities such as swallowing dysfunction, aspiration, lung disease, genetic syndromes, age-specific stratification, and follow-up duration were inconsistently reported or not reported at all. Twelve cases out of thirty-six cases regained the vocal cord mobility and were considered improved.

The HR for MUAP present versus absent was 5.56 (95% CI: 1.653 - 18.71), indicating that patients with MUAP present were 5.56 times

**Table 2**  
Pediatric LEMG for anatomical mapping and as an adjunct to therapeutic procedure.

| Author                        | Cases (n) | Age             | Pathology                                                       | Electrodes              | Summary of findings                                                                                                                                                                                                                                     | Risk of Bias |
|-------------------------------|-----------|-----------------|-----------------------------------------------------------------|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| Odegard et al., 2000 [25]     | 60        | Newborn to 16 y | Cases undergoing VATS for PDA                                   | IM (TA)                 | EMG of left RLN obtained in 98.3% cases                                                                                                                                                                                                                 | Low          |
| Caloway et al., 2020 [26]     | 3         | 2-14 y          | Unilateral VFI undergoing Ansa-RLN RLN re-innervation           | IM (TA, PCA)            | LEMG helped to alter the route of dissection in 62.7% cases                                                                                                                                                                                             | Low          |
| Zur and Carroll, 2015 [27]    | 33        | 2-16 y          | Unilateral VFI undergoing Ansa-RLN re-innervation               | IM (TA, PCA)            | LEMG helped to identify the neural pathway integrity based on amplitude and latency of muscle action potential, helping in clinical decision making for RLN-Ansa anastomosis                                                                            | Low          |
| Allen et al., 1991 [28]       | 1         | 9 y             | Hyperfunctional dysphonia                                       | Skin surface electrodes | LEMG was used to assess surgical candidacy; patients with normal EMG findings were deferred from Ansa-RLN anastomosis                                                                                                                                   | Low          |
| Fayoux et al., 2020 [29]      | 16        | 2 m-16 y        | Unilateral or Bilateral VCP undergoing re-innervation           | IM (NM)                 | EMG guided biofeedback training was useful in improving the dysphonia unresponsive to conventional voice therapy                                                                                                                                        | Medium       |
| Lee et al., 2020 [30]         | 8         | 2.2 to 18 y     | Bilateral VCP undergoing selective laryngeal innervation        | NM                      | LEMG was used preoperatively to assess denervation status and post-operatively to grade the recovery following reinnervation with NMP                                                                                                                   | Low          |
| Meyer et al., 2006 [31]       | 11        | 11-16 y         | Cases undergoing thyroidectomy                                  | IM(TA, CT)              | LEMG was used preoperatively which showed no function in five cases and paradoxical movement of PCA muscles in three cases                                                                                                                              | Medium       |
| St. Louis et al., 1982 [32]   | 1         | 4 y 3 m         | Case of stutter                                                 | Skin surface electrodes | RLN and EBSLN were consistently identified without complications, with stable intraoperative signals in all cases. One patient developed transient left vocal cord paresis post-thyroidectomy despite normal signals, which resolved within six months. | Low          |
| Faoury et al., 2019 [33]      | 1         | 6 y             | Unilateral VCP undergoing laryngeal innervation                 | NM                      | EMG can be a potential tool for biofeedback training in cases undergoing speech therapy as shown by some degree of improvement in this case                                                                                                             | Low          |
| White et al., 2009 [34]       | 5         | 11-17 y         | Cases undergoing branchial pouch and thyroid surgeries          | Surface endotracheal    | LEMG showed no chances of recovery (findings not mentioned), based on which intervention was decided                                                                                                                                                    | Low          |
| Bowe et al., 2019 [35]        | 1         | 5 y             | Unilateral VCP                                                  | IM (TA, PCA)            | LEMG was useful in identifying the RLN intraoperatively and predicting post-operative outcome based on intraoperative findings in once case with paretic right RLN                                                                                      | Low          |
| Labuschagne et al., 2021 [36] | 23        | 3-12 y          | Cases of seizure undergoing vagal nerve stimulator implantation | Surface endotracheal    | Progressive loss of signal in left TA and PCA muscle guided on opting for laryngeal re-innervation                                                                                                                                                      | Low          |
| Ongkasuwan et al., 2019 [37]  | 31        | 1.9 to 18 y     | Cases with Unilateral VCP Undergoing NSLR                       | IM (TA, PCA)            | IONM during VNS insertion helps optimize lead placement on CN 10, avoid non-vagal stimulation, identify variants, and detect injury, though voice outcomes remain unproven pending larger studies                                                       | Low          |
| Mikuni et al., 2004 [38]      | 2         | 11 y, 16 y      | Intracranial Astrocytoma, & Medulloblastoma undergoing surgery  | Surface endotracheal    | LEMG was used to confirm neuronal injury before NSLR, worse grade of pre-op LEMG scores co-related with improvement in phonation time and intensity                                                                                                     | Low          |

CN 10, Cranial nerve 10 (vagus nerve); CT, Crico-thyroid; EBSLN - External Branch of Superior Laryngeal Nerve; EMG, Electromyography; IM, Intramuscular; IONM, Intraoperative nerve monitoring; LEMG, Laryngeal electromyography; m, Months; NM, Not mentioned; NMP, Nerve Muscle Pedicle; NS, Not Significant; NSLR, Non-specified laryngeal resection; PCA, Posterior cricoarytenoid; PDA, Patent ductus arteriosus; RLN, Recurrent laryngeal nerve; TA, Thyroarytenoid; VATS, Video-assisted thoracoscopic surgery; VCP, Vocal cord paralysis; VFI, vocal fold immobility; VNS, Vagal nerve stimulator; y, Years.

**Table 3**  
Pediatric LEMG for developmental and physiological study.

| Author                       | Cases (n) | Age      | Pathology                                           | Electrodes                    | Summary of findings                                                                                                                                                                               | Risk of Bias |
|------------------------------|-----------|----------|-----------------------------------------------------|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| Eichenwald et al., 1992 [39] | 36        | Infants  | Full-term, premature infants with and without apnea | Esophageal surface electrodes | The typical coordination between PCA and diaphragm muscles is frequently disturbed in newborns but improves with development and aminophylline stimulation.                                       | Low          |
| Kosch et al., 1988 [40]      | 8         | 2-3 days | Full-term healthy infants                           | Esophageal surface electrodes | Expiratory braking in newborns involves coordinated diaphragm activity, reduced PCA activity, and increased laryngeal adduction, with pattern variations linked to respiratory system immaturity. | Medium       |

PCA - Posterior cricoarytenoid.

more likely to recover than those with MUAP absent. The p-value for the regression coefficient was 0.00558, which shows a statistically significant association between MUAP and recovery. The Concordance Index (C-index) was 0.714, indicating moderate discriminatory power of the model.

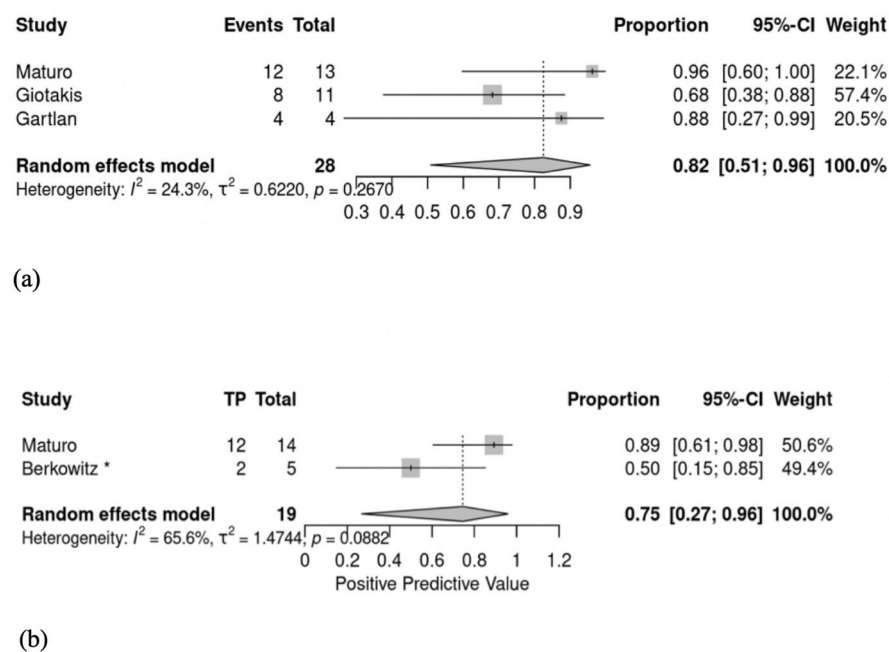
Similarly, the KM curve analysis demonstrated a significant difference in vocal cord recovery between patients with and without MUAP on LEMG ( $p = 0.00094$ ). Patients with MUAP (grey line) exhibited higher chances of recovery over time compared to those without MUAP (blue line). (Fig. 3(b)).

Despite the observed heterogeneity in etiology, laterality, and comorbid conditions, subgroup analyses could not be performed because of inconsistent and incomplete reporting of potential confounders and recovery-modifying factors across studies. Consequently, the influence of these additional risk factors on recovery could not be adequately assessed.

#### 4. Discussion

Given the potential role of LEMG in clinical laryngology and airway





**Fig. 2.** Forest plot showing sensitivity meta-analysis (a) and positive predictive value meta-analysis (b) (a) shows the Forest plot showing the pooled sensitivity of Laryngeal Electromyography for predicting recovery in pediatric vocal cord paralysis across three studies. The pooled sensitivity was 82% (95% CI: 51–96%), with individual study sensitivities ranging from 68% to 96% and low heterogeneity ( $I^2 = 24.3\%$ ) (b) shows Forest plot showing the pooled Positive Predictive Value of Laryngeal Electromyography across two studies. The pooled Positive Predictive Value was 75% (95% CI: 27–96%), with individual study Positive Predictive Values ranging from 50% to 89% and moderate-to-high heterogeneity ( $I^2 = 65.6\%$ ).

surgery yet with the lack of understanding of its utility and its relatively sparse application in the pediatric population compared to the adult population, in this work we have sought to comprehensively analyze the prevalent clinical applications and indications of LEMG in the pediatric population, with particular emphasis on its prognostic utility in predicting recovery from VCP.

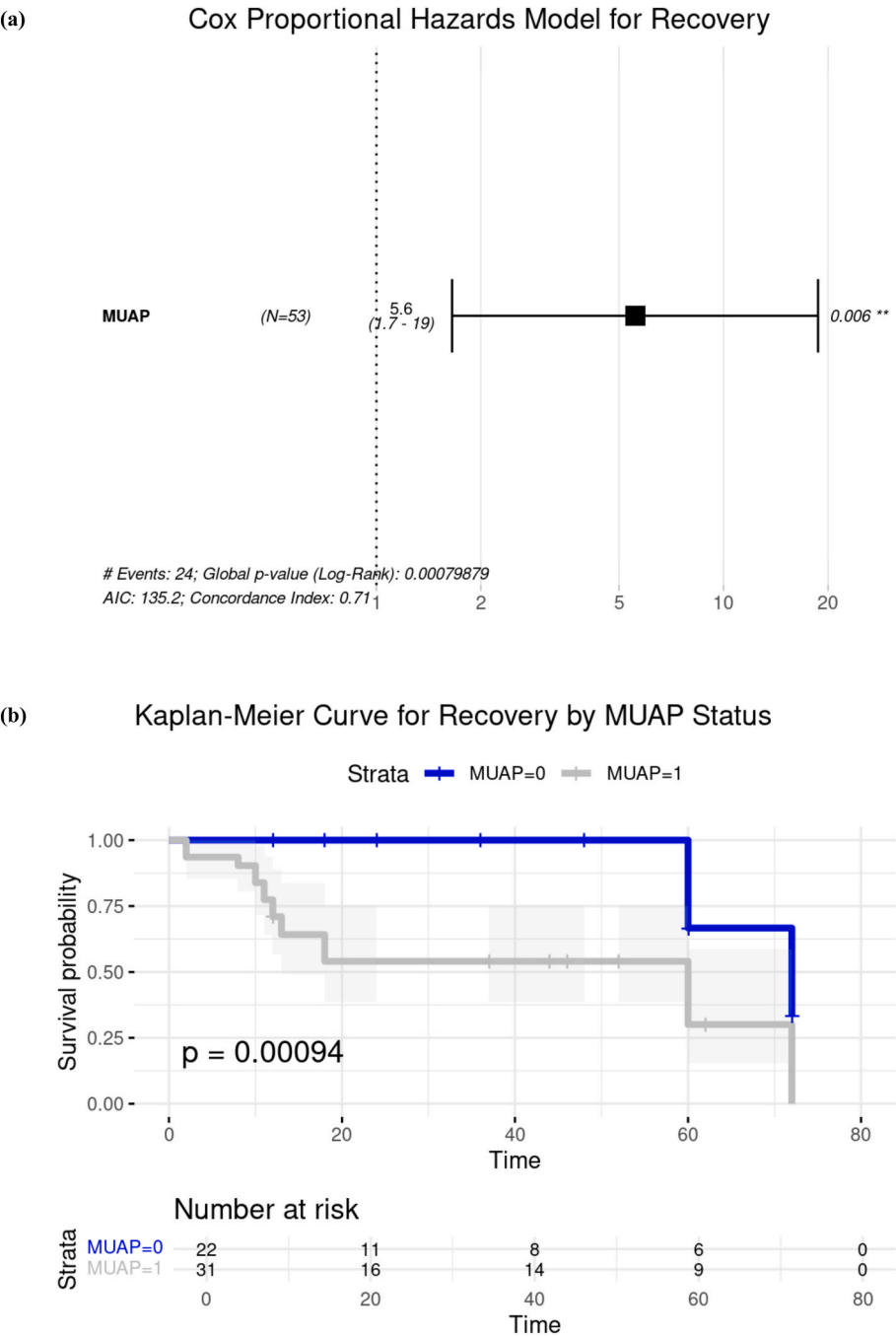
4.1. Electrodes

During LEMG, detecting neuromuscular activity in the relevant muscles requires placement of electrodes either directly within the muscle or at least in close proximity to it. Based on our review, electrode placement could be broadly classified as either intramuscular or surface (overlying the muscle). Intramuscular electrodes typically require anesthesia and allow measurement of electrical activity during volitional tasks such as coughing. The use of intramuscular electrodes was the most predominant approach, with applications ranging from diagnostic to prognostic purposes, as well as anatomical mapping of the RLN and vagus nerve. The thyroarytenoid muscle (TA) was the most commonly targeted site (in 21 out of 33 studies, 63.63%), followed by the posterior cricoarytenoid (PCA) (in 9 out of 33 studies, 27.27%) and cricothyroid (CT) (in 3 out of 33 studies, 9.09%) muscles. Surface electrodes, including endotracheal and esophageal types, were used to measure potentials from the TA and PCA muscles, respectively. While endotracheal electrodes were predominantly employed during surgeries involving risk to the RLN or vagus nerve for intraoperative nerve monitoring (IONM), we also identified studies using esophageal electrodes to explore the relationship between PCA activity and diaphragmatic function in neonatal respiratory physiology [39,40]. Additionally, two studies applied surface skin electrodes assessing CT muscle activity in cases of hyperfunctional dysphonia, where LEMG was successfully used as a biofeedback mechanism. [ 28, 32].

4.2. Indications

4.2.1. Diagnostic

One of the important applications of LEMG in the pediatric population lies in its diagnostic value, particularly in differentiating VFI caused by denervation from that due to mechanical fixation. VFI can arise either from a neurogenic cause or from mechanical factors such as cricoarytenoid joint fixation/PGS, or arytenoid dislocation. Although intra-operative palpation of the cricoarytenoid joint and assessment of its mobility may provide insights into fixation, such evaluation can often be subjective. Several studies, including those by Carrat et al., [8], Aragon-Ramos et al., [9], Wohl et al., [10], Jacobs and Finkel [11], Woo and Arnadia [12], Koch et al. [14], Ysunza et al. [15] and Scott et al., [17], have demonstrated the utility of LEMG in distinguishing mechanical obstruction from neurogenic VFI, thereby guiding treatment strategies. The key parameter they used to guide their decisions was the presence of MUAP which signified intact innervation. However, the presence of MUAP does not always imply that VFI is purely mechanical. It may also represent neurogenic VCP with a potential for recovery over time, as shown in studies by Giotakis & Potosching [18], Scott et al. [19], Berkowitz et al. [20] and Gartlan et al. [21]. Moreover, a study by Martinez et al. assessed cases of VFI with normal LEMG findings and found that etiologies such as idiopathic paralysis, central nervous system injury, and vagal mononeuritis could also present with normal results [41]. This highlights the importance of a judicious evaluation of VFI cases, emphasizing the need to correlate clinical findings rather than relying solely on LEMG outcomes before embarking on any surgical intervention. A particularly noteworthy but rarely reported finding was described by Berkowitz et al. [20]. Their study highlighted the importance of phasic activity, specifically, phasic activation of the TA during expiration and the PCA during inspiration. They observed that a child with this phasic pattern recovered, whereas another child with MUAPs that were



**Fig. 3.** Cox proportional hazards model for recovery (a) and Kaplan Meier Curve for recovery by Motor Unit Action Potential status (b) (a) illustrates the Cox proportional hazards regression model evaluating the association between presence of Motor Unit Action Potential on Laryngeal Electromyography and recovery in pediatric vocal cord paralysis across 4 studies. Patients with Motor Unit Action Potential present were 5.56 times more likely to recover compared to those without Motor Unit Action Potential (HR 5.56, 95% CI: 1.653–18.71,  $p = 0.00558$ ), indicating a statistically significant association. AIC; Akaike Information Criterion (b) Kaplan Meier curve illustrating recovery over time by Motor Unit Action Potential status ( $p = 0.00094$ ). Patients with Motor Unit Action Potential (grey line) had higher chances of recovery over time compared to those without Motor Unit Action Potential (blue line), demonstrating the predictive value of Motor Unit Action Potential for favorable recovery.

aberrant and non-phasic failed to recover. This observation underscores that not only the presence but also the quality and pattern of MUAP activity carry important prognostic implications.

A rare utility of LEMG was demonstrated in a case initially suspected to be VCP but later identified as congenital central hypoventilation syndrome (CCHVS). LEMG analysis revealed that the vocal cord adduction was synchronous with diaphragmatic pacing, leading to deferral of the VCP diagnosis. (24).

**4.2.2. Prognostic**

Prognosis of VCP remains a challenge even today, particularly in cases of idiopathic bilateral VCP. The optimal waiting period for spontaneous recovery has not been clearly defined, and the likelihood of recovery cannot be reliably predicted. This uncertainty often leaves parents confused when choosing appropriate treatment options for their child.

In our study, although based on a limited number of reports, the

calculated HR indicated that the probability of recovery was five times higher when MUAPs were present compared to when they were absent. The KM curve also demonstrated a statistically significant increase in recovery rates over a 12-month period in the presence of MUAPs. Furthermore, pooled analysis showed a sensitivity of 82% and a PPV of 75% for MUAPs in predicting recovery.

A retrospective review by Statham et al. employed the ratio of MUAP amplitude during sniff to that during phonation, using a cutoff value of 0.65 as electrophysiologic evidence of synkinesis in cases of VCP. The presence of synkinesis by this criterion yielded a sensitivity and NPV of 100%, highlighting synkinesis as a poor prognostic marker for recovery. They emphasized that laryngeal synkinesis should be suspected in patients who demonstrate good motor unit recruitment to task despite persistent VFI [42]. In contrast, Smith et al. [43] reported a higher PPV of 100% and an NPV of 89.5% for predicting recovery at six months when applying the LEMG-based prognostic categories (excellent, fair, and poor prognosis) defined by Munin et al., which incorporated recruitment patterns, MUAP configuration, fibrillation potentials, and synkinesis to stratify the prognosis [44]. Similarly, a meta-analysis by Rickert et al. evaluating 503 patients with VCP found that the pooled PPV for predicting poor recovery was 90.9%, while the NPV for predicting recovery was more modest at 55.6% [3]. It should be noted that these four studies [42–45] were performed in adult populations.

Based on the above findings, while our study demonstrated a moderate predictive value of LEMG for favorable prognosis when MUAPs were present, the studies by Rickert et al. and Statham et al. highlighted that the predictive ability of LEMG for poor prognosis is considerably higher.

#### 4.2.3. Anatomical mapping of RLN and Vagus nerve

LEMG is a well-established tool for identifying the RLN or vagus nerve during surgeries where these structures are at risk. Our qualitative analysis revealed diverse applications of LEMG, with all included studies demonstrating its benefit in accurately localizing the nerve of interest. The surgical procedures in which LEMG was employed included thyroid surgeries, branchial pouch excisions, intracranial tumor resections, and implantation of vagus nerve stimulators for intractable seizures (Table 2). The electrodes being employed were either intramuscular or surface endotracheal.

A meta-analysis by Liu et al., encompassing 1242 exposed RLNs, demonstrated that IONM significantly reduced the incidence of temporary RLN palsy and showed a nonsignificant trend toward reducing permanent palsy in patients undergoing endoscopic thyroid surgery [45]. In contrast, a meta-analysis evaluating its role in conventional thyroid surgery revealed no benefit over visual nerve identification [46]. However, another meta-analysis focusing on high-risk thyroidectomies reported a statistically significant reduction in RLN injury rates, showing a lower overall incidence (4.5% vs. 2.5%) as well as a lower rate of temporary palsy (3.9% vs. 2.4%) when compared to visual assessment [47]. These three studies [45–47] were performed in adult populations. These findings highlight the potential use of LEMG in anatomical mapping.

#### 4.2.4. Adjunct to therapeutic procedure

Beyond its diagnostic applications, LEMG as an adjunct to therapeutic procedures still has significant gaps in the literature. One of the most frequent therapeutic uses is to confirm laryngeal denervation prior to undertaking reinnervation surgery. Since this procedure requires transection of the RLN and its anastomosis with other functional nerves such as the ansa cervicalis, verifying neural integrity is essential. In our analysis, six studies reported the use and benefit of LEMG in this context. Furthermore, although rare, case reports of two patients, one with stuttering and another with hyperfunctional dysphonia, demonstrated benefit from LEMG-guided biofeedback. A study by Stemple et al. demonstrated that LEMG biofeedback could reduce laryngeal muscle tension in patients with vocal nodules, this study was performed in an

adult population [48]. Another noteworthy therapeutic application is in guiding botulinum toxin injection, however, we did not identify any reports of its use in the pediatric population. Based on their review, Sataloff et al. have provided a level C recommendation, suggesting LEMG to be possibly useful or predictive for guiding botulinum toxin injections into the TA muscle in cases of spasmodic dysphonia, but the evidence is not strong enough to be considered definitive, and further high-quality studies are required [4].

#### 4.2.5. Basic science research

Another important application of LEMG lies in basic science research, particularly in elucidating the phasic relationships between laryngeal muscle activity and the various phases of respiration. Two studies employed esophageal electrodes to evaluate PCA muscle activity. Eichenwald et al. demonstrated that in premature infants, the normal sequence of PCA activation followed by diaphragmatic contraction is disrupted, but improves with maturation and administration of aminophylline. (39) Similarly, Kosch et al. described an expiratory braking mechanism, whereby coordinated diaphragmatic activity and reciprocal PCA activation help regulate end-expiratory lung volume and ventilation [40]. This study [40] was performed in neonates.

#### 4.3. Limitations

This study has several limitations. Although it provides a qualitative review of the various indications for LEMG use in the pediatric population, it does not offer specific recommendations for its use in pediatric otolaryngology. Nevertheless, the reported indications demonstrate clear benefits in the pathologies studied. Only a very limited number of studies were included in the quantitative analysis for predicting recovery. This analysis is constrained by factors such as disease course heterogeneity, inconsistent reporting parameters, and small sample sizes, resulting in wide CIs. Additionally, given the variability in injury duration and stages of recovery, our analysis cannot serve as a complete predictor of recovery, as the European Laryngological Society has documented diverse MUAP patterns across different recovery stages [49].

These limitations highlight the need for standardized, protocol driven reporting of LEMG findings. Such reporting should involve homogeneous populations with similar etiologies, demographic characteristics, disease duration, and stages of recovery to improve the reliability and comparability of future studies.

#### 5. Conclusion

LEMG in pediatric laryngology and airway surgery serves as a valuable diagnostic, therapeutic, and prognostic adjunct with diverse applications. It shows moderate accuracy in predicting favorable recovery in cases of VCP when normal MUAPs are present, thereby aiding clinical decision-making. Additionally, LEMG is useful for differentiating mechanical from neurogenic VFI, guiding reinnervation procedures, and facilitating anatomical mapping of the RLN and vagus nerves during complex surgeries.

#### CRedit authorship contribution statement

**Bigyan Raj Gyawali:** Writing – review & editing, Software, Project administration, Methodology, Formal analysis, Conceptualization. **Sonam Dhenga:** Writing – original draft, Software, Resources, Data curation. **Simran Rauniyar:** Writing – original draft, Software, Resources, Data curation. **Sony KC:** Writing – original draft, Resources, Data curation. **Milan Pokhrel:** Writing – original draft, Software, Resources, Formal analysis, Data curation.



## Human rights committee statement

This study is a systematic review and meta-analysis of published literature and did not involve direct patient contact. Therefore, it was exempt from institutional human subjects review.

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## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## List of abbreviations

| Abbreviation   | Full Forms                                                         |
|----------------|--------------------------------------------------------------------|
| AIC            | Akaike Information Criterion                                       |
| CA             | Cricothyroid                                                       |
| CCHVS          | Congenital central hypoventilation syndrome                        |
| CI             | Confidence interval                                                |
| CN 10          | Cranial nerve 10 (vagus nerve)                                     |
| CT             | Cricothyroid                                                       |
| DOR            | Diagnostic odds ratio                                              |
| EBSLN          | External branch of superior laryngeal nerve                        |
| EMG            | Electromyography                                                   |
| FN             | False negative                                                     |
| FP             | False positive                                                     |
| HR             | Hazard ratio                                                       |
| I <sup>2</sup> | Higgins' I-squared heterogeneity statistic                         |
| IM             | Intramuscular                                                      |
| IONM           | Intraoperative nerve monitoring                                    |
| JB             | Joanna Briggs Institute                                            |
| KM             | Kaplan Meier                                                       |
| LEMG           | Laryngeal electromyography                                         |
| MUAP           | Motor unit action potential                                        |
| NM             | Not mentioned                                                      |
| NMP            | Nerve muscle pedicle                                               |
| NOS            | Newcastle–Ottawa Scale                                             |
| NPV            | Negative predictive value                                          |
| NS             | Not specified                                                      |
| NSLR           | Non-specified laryngeal resection                                  |
| PCA            | Posterior cricoarytenoid                                           |
| PDA            | Patent ductus arteriosus                                           |
| PGS            | Posterior glottic stenosis                                         |
| PPV            | Positive predictive value                                          |
| PRISMA         | Preferred Reporting Items for Systematic Reviews and Meta-Analyses |
| RLN            | Recurrent laryngeal nerve                                          |
| SMU            | Single motor unit                                                  |
| TA             | Thyroarytenoid                                                     |
| TN             | True negative                                                      |
| TP             | True positive                                                      |
| VATS           | Video-assisted thoracoscopic surgery                               |
| VCP            | Vocal cord paralysis                                               |
| VFI            | Vocal fold immobility                                              |
| VNS            | Vagal nerve stimulator                                             |

## Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ijporl.2026.112781>.

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